MIL-T-85298(AS)
15 January 1979
Superseding MIL-T-81134
Dated 21 January 1976

MILITARY SPECIFICATION

TEST SET, WEAPON CONTROL, AIRCRAFT AN/ASM-184B(V)1 AND AN/ASM-184B(V)2

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The test sets covered by this specification are for hanger deck and flight line use and perform operational and safety tests on aircraft wiring and video circuits associated with Walleye Weapons.

1.2 Classification - The equipment covered by this specification shall consist of the items listed in paragraph 1.2.1. (Navy) or paragraph 1.2.2 (Air Force).

1.2.1 Aircraft Weapon Control Test Set AN/ASM-184B(V)1 - This test set shall consist of the following items

Item	Type Designation	Paragraph
Aircraft Weapon Control Test Set	TS-2475B/ASM-184(V)	3.5.1
Electrical Dummy Load	DA-514/ASM-184(V)	3.5.2
Connector Adapter (two required)	MX-8311/ASM-184(V)	3.5.3
Test Set Case	CY-6643/ASM-184(V)	3.5.5
Electrical Power Cable	W1 of AN ASM-184B(V)	3.5.6
Service Cable	W2 of AN/ASM-184B(V)	3.5.7

1.2.2 Aircraft Weapon Control Test Set AN/ASM-184B(V)2 - This test set shall consist of the following items

Item	Type Designation	Paragraph
Aircraft Weapon Control Test Set	TS-2475B/ASM-184(V)	3.5.1
Electrical Dummy Load	DA-514/ASM-184(V)	3.5.2
Connector Adapter (2 required)	MX-8312/ASM-184(V)	3.5.4
Test Set Case	CY-6643/ASM-184(V)	3.5.5
Electrical Power Cable	W1 of AN ASM-184B(V)	3.5.6
Service Cable	W2 of AN/ASM-184B(V)	3.5.7

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

FSC-6625

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-7788	Plate, Plastic Lighting
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution

DRAWINGS

Naval Air Systems Command

DL 64A15F2002	Data List for Aircraft Weapon Control Test Set AN/ASM-184B(V) and all drawings therein

PUBLICATIONS

Military

ANA Bulletin No. 400	Electronic Equipment, Aircraft and Guided Missiles, Applicable Documents

Naval Air Systems Command

WR-43	Preparation of Quality Assurance Provisions

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2.1.1 Availability of Documents -

(a) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications and documents required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provisions for preproduction testing.

3.2 Parts and Materials -

3.2.1 General Requirements - The parts and materials specified in the drawings and related documents for use in construction of this equipment were selected to ensure the fulfillment of all requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change and the cost associated therewith upon the contractor. Approval for deviations must be requested from the procuring activity by the contractor and granted by the procuring activity prior to submission of equipments for preproduction tests. Approval of parts and material deviations shall not be construed by the contractor to constitute a waiver of the performance requirements or any other requirements of this specification.

3.2.1.1 Basic Conformance - Parts and materials specified for use in the equipment were selected to conform to the requirements of MIL-T-21200. Parts were selected from parts as specified in ANA Bulletin No. 400, unless the parts listed therein would not perform adequately in the intended applications and environments specified herein.

3.2.1.2 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained by the contractor as specified in MIL-T-21200.

3.2.1.3 Interchangeability - For guidance in establishing interchangeability, table I lists nonrepairable subassemblies used in the test set.

3.3 Design and Construction - The equipment shall be manufactured in accordance with the drawings and subsidiary specifications listed in D. 64A15F2002 and related lists, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this and related specifications. In any case it is a condition of final acceptance that the equipment manufactured meet all the performance requirements of this specification, and such acceptance test documents as may be applicable. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be specifically approved by the procuring activity prior to their incorporation.

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TABLE I. Nonrepairable subassemblies used in the test set

Subassembly	Drawing No.
Extender Board Assembly (A3A1)	64A15F1261
Voltage Regulator Assembly (A3A3)	64A15F1264 or 64A15C1400
Staircase Generator Assembly (A3A4) (Except for Quartz Crystal Unit 65A7A48-1)	64A15F1265 or 64A15C1420
Divider Assembly (A3A5)	64A15F1266 or 64A15C1440
Pulse Generator Assembly (A3A6)	64A15F1267 or 64A15C1460
Video Switching Assembly (A3A7)	64A15F1268
Composite Video Assembly (A3A8)	64A15F1369
Moving Band Generator Assembly (A3A9)	64A15F1270
Resolution Line Generator Assembly (A3A10)	64A15F1271
Test Set Case (CY-6643/ASM-184(V) (Except for Clamping Catch ZSP2-2041)	64A15F2085
Electrical Power Cable (W1 of AN/ASM-184B(V)	64A15F1063

3.3.1 Total Weight - The total weight of the equipment, including cables, shall not exceed 43.0 pounds (19.5 kilograms).

3.3.2 Form Factor - With the test sets assembled in their carrying cases, the overall dimensions of the test sets (excluding handles and latches) shall not exceed 12.00 by 12.00 by 17.50 inches (305 x 305 x 445 millimeters).

3.3.3 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.3.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 200 hours (one MTBF) without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.3.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with approved servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and the wearout life of such parts shall be determined by the contractor and submitted to the procuring agency.

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3.3.3.3 Reliability in Specified Mean-Time-Between-Failure (MTBF) - The specified MTBF of the equipment (θ_0 as defined in MIL-STD-781) shall be 200 hours when tested and accepted as outlined under the requrements of 4.9.4 and 4.9.5.

3.3.4 Cabling and Connections -

3.3.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with specification MIL-E-5400.

3.3.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of specification MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference and vulnerability of the equipment to radio interference shall be within limits conforming to MIL-I-6181 with the following deviations

3.3.6.1 Short Duration Interference Limits - Limits for short duration interference shall be extended as follows

Switch	Maximum Duration	MIL-I-6181 Limit Extension
FUNCTION (A5A1S1)	1 second	+ 20 db
POWER (A5S1)	3 seconds	No upper limit

3.3.6.2 Steady State Broadband Radiated Limits - Limits for steady state broadband radiated emission shall be extended as follows

Frequency	MIL-I-6181 Limit Extension
2.0 to 4.0 MHz	6 db
4.1 to 5.0 MHz	15 db
12.0 to 12.5 MHz	15 db
121.0 to 123.0 MHz	10 db

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3.3.7 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (25 ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input power voltage	Three phase, 115 ± 1.0 volts ac, 400 ± 5 Hz

3.3.8 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for class 2 equipment, except as modified herein.

3.3.9 Warm Up Time - The time required for the equipment to warm-up prior to operations shall not exceed 1 minute under standard conditions and 5 minutes at extreme service conditions.

3.3.10 Primary Input Power Requirements - The equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in MIL-T-21200. The power required shall be of the following type and shall not exceed the specified amounts:

AC Power, Three Phase, 4 Wire 115 ± 11.5 Volts, 400 ± 20 Hz, at 300 VA. Phase rotation ABC on pins 23, 22, and 21 (pin 7 neutral) respectively of connector J1.

3.3.10.1 Protection - The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as specified in MIL-STD-704 for Category B equipment.

3.4 Performance - The performance of the equipment shall be as set forth below plus any and all performance requirements stated on the drawings of DL 64A15F2002. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions.

3.5 Detail Requirements - Aircraft Weapon Control Test Sets AN/ASM-184B(V)1 and AN/ASM-184B(V)2 shall be manufactured in accordance with the documents listed on DL 64A15F2002 and shall meet the requirements stated thereon and in the following subparagraphs.

3.5.1 Aircraft Weapon Control Test Set TS-2475B/ASM-184(V) - The test set shall be capable of meeting the following detail requirements.

3.5.1.1 Voltage Measurements - The test set shall measure and register the individual voltage quantities as applied to the specified pins of connector J1. The test set shall be capable of performing the following self-tests and

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aircraft checks with the FUNCTION switch in the positions indicated. All test requirements are performed with the PUSH-TEST switch depressed.

3.5.1.1.1 Switch Position 1, Aircraft Phase A Power - With primary input power applied, the test set shall measure the aircraft phase A power, 115 ± 11.5 volts rms, 400 ± 20 Hz between pins 23 and 7 of J1. An input voltage of 103.5 ± 2.5 volts rms, 400 ± 20 Hz shall cause the QUALITY meter to register over the left edge of the area containing a triangle. A voltage of 126.5 ± 2.5 volts rms, 400 ± Hz shall cause the meter to register over the right edge of the area containing a triangle.

3.5.1.1.2 Switch Position 2, Aircraft Phase B Power - With primary input power applied, the test set shall measure the aircraft phase B power, 115 ± 11.5 volts rms, 400 ± 20 Hz between pins 22 and 7 of J1. An input voltage of 103.5 ± 2.5 volts rms, 400 ± 20 Hz shall cause the QUALIY meter to register over the left edge of the area containing a triangle. A voltage of 126.5 ± 2.5 volts rms, 400 ± 20 Hz shall cause the meter to register over the right edge of the area containing a triangle.

3.5.1.1.3 Switch Position 3, Aircraft Phase C Power - With primary input power applied, the test set shall measure the aircraft phase C power, 115 ± 11.5 volts rms, 400 ± 20 Hz between pins 21 and 7 of J1. An input voltage of 103.5 ± 2.5 volts rms, 400 ± 20 Hz shall cause the QUALITY meter to register over the left edge of the area containing a triangle. A voltage of 126.5 ± 2.5 volts rms, 400 ± 20 Hz shall cause the meter to register over the right edge of the area containing a triangle.

3.5.1.1.4 Switch Position 4, Aircraft Input Power Phase Sequence - With primary input power applied, the test set shall measure the phase sequence of the aircraft three-phase input power. With phases ABC connected to pins 23, 22, and 21 (pin 7 neutral) respectively of J1, the QUALITY meter shall register in the area containing a triangle. With any two phases interchanged, the meter shall register to the left of the area containing a triangle.

3.5.1.1.5 Switch Position 5, Guidance Section Energize Signal - The test set shall measure the guidance energize signal, 28 ± 4.2 volts dc between pins 20 (positive) and 7 (ground) of J1. An input signal of 23.8 ± 0.5 volts dc shall cause the QUALITY meter to register over the left edge of the area containing a circle. A voltage of 32.2 ± 0.5 volts dc shall cause the meter to register over the right edge of the area containing a circle.

3.5.1.1.6 Switch Position 6, 28-Volt DC Power Supply (self-test) - The test set shall measure the output voltage of the 28-volt dc power supply contained in the set test. The output signal of the 28-volt dc power supply shall cause the QUALITY meter to register within the area containing a circle.

3.5.1.1.7 Switch Position 7, Composite Video Circuits (self-test) - The test set shall measure the composite video peak holder signal, developed by the composite video peak holder circuits of the video generator assembly. This signal shall cause the QUALITY meter to register within the area containing a circle.

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3.5.1.1.8 Switch Position 8, Horizontal Sync Circuits (self-test) - The test set shall measure the horizontal sync peak holder signal, developed by the horizontal sync peak holder circuits of the video generator assembly. This signal shall cause the QUALITY meter to register within the area containing a circle.

3.5.1.1.9 Switch Position 9, Vertical Sync Circuits Check (self-test) - The test set shall measure the vertical sync peak holder signal, developed by the vertical sync peak holder circuits of the video generator assembly. This signal shall cause the QUALITY meter to register within the area containing a circle.

3.5.1.1.10 Switch Position 10, Mechanical Uncage Signal - The test set shall measure the mechanical uncage signal, 28 ± 4.2 volts dc between pins 19 (positive) and 7 (ground) of J1. An input voltage of 23.8 ± 0.5 volts dc shall cause the QUALITY meter to register over the left edge of the area containing a circle. A voltage of 32.2 ± 0.5 volts dc shall cause the meter to register over the right edge of the area containing a circle.

3.5.1.1.11 Switch Position 11, Trigger Signal - The test set shall measure the trigger signal, 28 ± 4.2 volts dc between pins 18 (positive) and 7 (ground) of J1. An input signal of 23.8 ± 0.5 volts dc shall cause the QUALITY meter to register over the left edge of the area containing a circle. A voltage of 32.2 ± 0.5 volts dc shall cause the meter to register over the right edge of the area containing a circle.

3.5.1.1.12 Switch Position 12, Fuze Signal - The test set shall measure the firing fuze signal, +300 ± 6 volts dc and the firing fuze signal, -300 ± 6 volts dc between pins 17 and 7 of J1. An input voltage of plus or minus 294 ± 2.0 volts dc shall cause the QUALITY meter to register over the left edge of the area designated by an "X". A voltage of plus or minus 306 ± 2.0 volts dc shall cause the meter to register over the right edge of the area designated by an "X".

3.5.1.1.13 Switch Position 13, Fuze Signal - The test set shall measure the firing fuze signal, +195 ± 4 volts dc and the firing fuze signal, -195 ± 4 volts dc between pins 17 and 7 of J1. An input voltage of plus or minus 191 ± 1.5 volts dc shall cause the QUALITY meter to register over the left edge of the area designated by an "X". A voltage of plus or minus 199 ± 1.5 volts dc shall cause the meter to register over the right edge of the area designated by an "X".

3.5.1.1.14 Switch Position 14, Stray Voltage Signals on Fuze Circuits - An input voltage of 0.100 (+0.010, -0.000) volt rms at 400 ± 20 Hz or 0.100 (+0.010, -0.000) volt dc applied between pins 17 (positive) and 7 (ground) of J1 shall cause the QUALITY meter to register in the BAD area immediately adjacent to the area containing a rectangle.

3.5.1.1.15 Switch Position 15, Stray Voltage Signals, Intent-to-Launch Circuits - An input voltage of 0.100 (+0.010, -0.000) volt rms at 400 ± 20 Hz or 0.100 (+0.010, -0.000) volt dc applied between pins 15 (positive) and 7 (ground) of J1 shall cause the QUALITY meter to register in the BAD area immediately adjacent to the area containing the rectangle. The same voltage inputs applied

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to pins 16 and 7 (ground), and with the ITL-CRAB SHORT TEST switch depressed, shall cause the QUALITY meter to indicate in the BAD area immediately adjacent to the area containing the rectangle.

3.5.1.1.16 Switch Position 16, Intent-to-Launch Signal - The test set shall indicate the presence of an intent-to-launch signal on the proper circuit with an input signal of 28 ± 1.0 volts dc applied between pins 16 (positive) and 7 (ground) of J1. The QUALITY meter shall indicate in the area containing the circle.

3.5.1.1.17 Switch Position 17, Guidance Section Ground - The test set shall indicate the presence of a ground applied to pin 14 of J1. A ground applied to pin 14 shall cause the QUALITY meter to register in the center of the area containing a circle.

3.5.1.1.18 Switch Position 18, Control Section Ground - The test set shall indicate the presence of the control section ground applied to pin 13 of J1. A ground applied to pin 13 shall cause the QUALITY meter to register in the center of the area containing a circle.

3.5.1.1.19 Switch Position 19, Caging Retainer and Boresighter (CRAB) Ground and SHORT TEST - The test set shall indicate the presence of the CRAB ground (desired condition) and a possible short (not desired) on the CRAB Aircraft circuit. A ground applied to pin 4 of J1 shall cause the QUALITY meter to indicate in the center of the area containing the circle. A resistance of 45K ± 5K applied to pin 4 and ground shall cause the QUALITY meter to indicate in the rectangle area with the ITL-CRAB SHORT TEST switch depressed.

3.5.1.1.20 Switch Position 20, Guidance Delay Ground - The test set shall indicate the presence of the guidance delay ground applied to pin 9 of J1. A ground applied to pin 9 shall cause the QUALITY meter to register in the center of the area containing a circle.

3.5.1.1.21 Switch Position 21, Negative DC Stray Voltage Signal, Fuze Circuits - An input voltage of 0.100 (+0.010, -0.000) volt dc applied between pins 17 (negative) and 7 (ground) of J1 shall cause the QUALITY meter to register in the BAD area immediately adjacent to the area containing a rectangle.

3.5.1.1.22 Switch Position 22, Negative DC Stray Voltage Signal, Intent-to-Launch Circuits - An input voltage of 0.100 (+0.010, -0.000) volt dc applied between pins 15 (negative) and 7 (ground) of J1 shall cause the QUALITY meter to register in the BAD area immediately adjacent to the area containing a rectangle.

3.5.1.1.23 Switch Position 23, Camera Energize Signal - The test set shall measure the camera energize signal, 28.0 ± 4.2 volts dc applied between pin 10 or 11 (positive) and pin 7 (ground) of J1. An input voltage of 23.8 ± 0.5 volts dc shall cause the QUALITY meter to register over the left edge of the area containing a circle. A voltage of 32.2 ± 0.5 volts dc shall cause the meter to register over the right edge of the area containing a circle.

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3.5.1.2 Grounds - The test set shall supply the following grounds.

3.5.1.2.1 The test set shall provide a ground at pins 5, 6, and 7 of J1 independent of the FUNCTION switch position.

3.5.1.2.2 The test set shall provide a ground, resistance of 0.5 ohm or less, at pin 30 of J1 for all FUNCTION switch positions except OFF and positions 7 through 11. Pin 30 of J1 shall be connected to the positive lead of the measuring device.

3.5.1.2.3 The resistance between pin 30 of J1 and ground with the FUNCTION switch in the OFF position shall be no less than 5.0 megohms.

3.5.1.2.4 The resistance between pin 30 of J1 and ground with the FUNCTION switch in positions 7 through 11 shall be less than 30 ohms.

3.5.1.3 POWER Lamp, Front Panel Illumination, and HOURS Meter - The test set front panel and POWER lamp shall illuminate and the HOURS meter shall operate when the POWER switch is in the ON position.

3.5.1.4 Video Generator Output Signals - The video generator shall produce a composite video signal output to be used in checking the TV display in the aircraft. There shall be three modes of operation with options within each mode as detailed below. However, regardless of the operating mode or option, all pulses generated by the video generator shall be controlled, with respect to time of occurrence, by a single, crystal-controlled master oscillator. The frequency of the master oscillator shall be 126.000 KHz ± 0.02 percent.

3.5.1.4.1 Crossbar Pattern - The signal produced by the video generator in this mode shall appear on the aircraft TV display as a crossbar pattern, seven blocks in the vertical plane and five plus partial blocks on each side in the horizontal plane. The signal shall also produce straight black horizontaland vertical grid lines which separate each of the blocks one from another. Each of the blocks shall contain one shade of gray corresponding to a step in a logarithmic gray scale. A total of seven steps corresponding to seven shades of gray shall be produced by the video generator. The two options within this mode shall be:

(a) Normal amplitude shall be displayed when the FUNCTION switch is in positions 1 and 12 through 22.

(b) Reduced amplitude shall be 50 ± 10 percent normal amplitude and shall be displayed when the FUNCTION switch is in position 23.

3.5.1.4.2 Resolution Lines - The signal produced by the video generator in this mode shall appear on the aircraft TV display as alternating black and white vertical displayed lines. The five options within this mode shall be.

(a) 200 lines resolution and shall be displayed when the FUNCTION switch is in position 2.

(b) 300 lines resolution and shall be displayed when the FUNCTION switch is in position 3.

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(c) 400 lines resolution and shall be displayed when the FUNCTION switch is in position 4.

(d) 500 lines resolution and shall be displayed when the FUNCTION switch is in position 5.

(e) 600 lines resolution and shall be displayed when the FUNCTION switch is in position 6.

3.5.1.4.3 Moving Band - The signal produced by the video generator in this mode shall appear on the aircraft TV display as a segment of resolution lines moving from left to right and may pause temporarily at each or either end of travel. The five options in this mode shall be

(a) 200 lines resolution moving band shall be displayed when the FUNCTION switch is in position 7.

(b) 300 lines resolution moving band shall be displayed when the FUNCTION switch is in position 8.

(c) 400 lines resolution moving band shall be displayed when the FUNCTION switch is in position 9.

(d) 500 lines resolution moving band shall be displayed when the FUNCTION switch is in position 10.

(e) 600 lines resolution moving band shall be displayed when the FUNCTION switch is in positions 11 and OFF.

3.5.1.4.4 Composite Video - The composite video signal output shall consist of horizontal sync pulses, vertical sync pulses, and synthetic video. The type of synthetic video displayed shall be determined by the FUNCTION switch position as specified in 3.5.1.4.1 through 3.5.1.4.3. The composite video signal output shall meet the following requirements

(a) HORIZONTAL SYNC PULSE

(1)	Repetition rate	15,750 Hz ± 0.02 percent
(2)	Duration	8.0 (+1.0, -1.5) microseconds
(3)	Blanking	16 (+1.0, -2.0) microseconds, including sync pulse
(4)	Amplitude	0.80 (+0.10, -0.25) volt peak-to-peak
(5)	Risetime and falltime	500 nanoseconds maximum
(6)	Polarity	Negative

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(7)	Interlace accuracy	Pulses of successive fields to be displaced by 1/2 line time ± 10 percent
(8)	Droop	0.05 volt maximum
(9)	Overshoot and undershoot	0.08 volt maximum
(b)	VERTICAL SYNC PULSE	
(1)	Repetition rate	60 Hz ± 0.02 percent
(2)	Duration	189 ± 19 microseconds
(3)	Blanking	
[a]	Rest (front porch)	Equal to three horizontal lines ± 10 percent. There shall be no horizontal sync pulses during front porch.
[b]	Blanking (back porch)	Equal to 10 horizontal lines ± 10 percent. Horizontal sync pulses to be generated during this interval.
(4)	Amplitude	0.80 (+0.10, -0.25) volt peak-to-peak
(5)	Risetime and falltime	1 microsecond maximum
(6)	Polarity	Negative
(7)	Droop	0.10 volt maximum
(8)	Overshoot and undershoot	0.08 volt maximum
(c)	SYNTHETIC VIDEO, CROSSBAR	
(1)	Amplitude	2.20 ± 0.50 volts peak-to-peak (black to white level)
(2)	Polarity	Positive (white level)
(3)	Voltage steps	7
(4)	Voltage difference (between successive steps)	0.3 ± 0.1 volt

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(5)	Voltage level, first step, ascending staircase	0.125 + 0.075 volt above black level
(6)	Step voltage rise and falltime	400 nanoseconds maximum
(7)	Step voltage overshoot	0.08 volt maximum into the sync and 0.12 volt maximum in the vertical direction
(8)	Step voltage droop	0.05 volt maximum
(9)	Vertical grid line pulses	
	[a] Duration	1.0 ± 0.25 microsecond
	[b] Overshoot	0.08 volt maximum into sync and 0.12 maximum in vertical direction
	[c] Risetime and falltime	400 nanoseconds maximum
(10)	Horizontal grid line pulses	
	[a] Duration	260 (+30, -60) microseconds
	[b] Overshoot	0.08 volt maximum

(d) SYNTHETIC VIDEO, RESOLUTION LINES

(1)	Amplitude	2.20 ± 0.50 volts peak-to-peak (black to white level)
(2)	Polarity	Positive
(3)	Resolution Lines	
	[a] 200 lines	2.0 MHz ± 10 percent
	[b] 300 lines	3.0 MHz ± 10 percent
	[c] 400 lines	4.0 MHz ± 10 percent
	[d] 500 lines	5.0 MHz ± 8 percent
	[e] 600 lines	6.0 MHz ± 8 percent
	[f] Pulse overshoot	0.08 volt maximum

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(e) SYNTHETIC VIDEO, MOVING BAND

(1)	Amplitude	2.20 + 0.50 volts peak-to-peak (black to white)
(2)	Polarity	Positive
(3)	Moving Band	
	[a] Width	10 ± 2 microseconds
	[b] Start	18 (+2, -1) microseconds from the leading edge of the horizontal sync pulses
	[c] Run length	31 ± 2.5 microseconds
	[d] Run duration	2 ± 0.3 seconds
(4)	Pulse overshoot	0.10 volt maximum

3.5.2 Electrical Dummy Load DA-514/ASM-184(V) - The electrical dummy load shall be capable of meeting the following detail requirements.

3.5.2.1 With circuit breakers CB1 or CB2 set to the ON position and 28 volts d-c applied in series with a 5-ohm resistor to pins A (positive) and B (negative) of J1 or J2, CB1 or CB2 shall open and lamp DS1 or DS2 shall light within 1 second after the application of the 28 volt power. At the same time, the voltage across the 5-ohm resistor shall drop to 0.2 volts or less.

3.5.3 Connector Adapter MX-8311/ASM-184(V) - Two Connector Adapters MX-8311/ASM-184(V) shall connect Electrical Dummy Load DA-514/ASM-184(V) of Aircraft Weapon Control Test Set AN/ASM-184B(V)1 to the ejector rack breech caps on Naval aircraft.

3.5.4 Connector Adapters MX-8312/ASM-184(V) - Two Connector Adapters MX-8312/ASM-184(V) shall connect Electrical Dummy Load DA-514/ASM-184(V) of Aircraft Weapon Control Test Set AN/ASM-184B(V)2 to the ejector rack breech assemblies on Air Force aircraft.

3.5.5 Test Set Case CY-6643/ASM-184(V) - The test set case shall house all components of the test set.

3.5.6 Electrical Power Cable W1 of AN/ASM-184B(V) - Cable W1 shall connect the test set to the WALLEYE weapon umbilical plug or the aircraft weapon sation tested and shall provide all power and test inputs to the test set during an aircraft checkout.

3.5.7 Service Cable W2 of AN/ASM-184B(V) - Cable W2 shall connect bench power sources to the test set and the video generator output to an oscilloscope or video monitor for test set calibration or repair.

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3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see 6.3).

3.7 Workmanship Defects - The contractor is responsible for assuring that discrepancies existing between hardware manufactured to the drawings and tested to the specification are not the result of workmanship defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspections - Test sets covered by this specification shall be subjected to the following inspections to determine compliance with all applicable requirements.

(a) Preproduction inspection (paragraph 4.6)

(b) Quality conformance inspection (paragraph 4.7)

(c) Periodic production inspection (paragraph 4.8)

4.3 Sampling -

4.3.1 Preproduction Sample - A preproduction sample of test sets manufactured in accordance with this specification shall be subjected to the preproduction tests detailed in 4.6 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the test sets by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic Production Samples - Periodic production samples shall be selected from test sets that have satisfactorily passed the quality conformance inspections of 4.7. These samples shall be subjected to the tests of 4.8 at an activity designated by the procuring activity. Periodic production sampling shall be as follows

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
For each additional 200 or fraction thereof	1

4.3.2.1 When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall

(a) Determine the cause of the defect.

(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

(d) Return to one sample out of the first 10 and proceed with periodic production sampling as specified in 4.3.2 when inspection is resumed.

4.4 Classification of Characteristics - The characteristics verified by the tests and examinations herein are classified as Critical, Major, or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.5 Test Procedures, Provisions, and Equipment -

4.5.1 Test Equipment - Unless otherwise specified in the contract or purchase order, provision of test equipment, including any necessary design, shall be the responsibility of the supplier. In addition to any test equipment required for completion of any tests or inspections required by referenced documents the following test equipment is required

(a) DC Power Supply with the following characteristics.

(1) Output voltage: Varible in the range of 190 to 320 volts dc

(2) Current requirement (maximum). 100 milliamperes at 320 volts dc

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(3) Ripple. Not greater than 1 millivolt

(4) Line and load regulation (combined). Not more than 4 millivolts.

(b) D-C Power Supply with the following characteristics:

(1) Output voltage Variable from zero to 36 volts dc

(2) Current requirement (maximum) 5 amperes at 36 volts dc

(3) Ripple Not greater than 40 Millivolts

(4) Line and load regulation (combined) Not more than 0.05 percent

(c) Ohmmeter with the following characteristics

(1) Range Zero to 5 megohms

(2) Accuracy 5 percent or better

(d) Digital Voltmeter with the following characteristics

(1) Input impedance 10 megohms with no more than a 20 picofarad shunt

(2) Range 0.000 to 999.9 volts dc and ac

(3) Sample rate Variable from 5 samples per second to 1 sample per 5 seconds

(4) Accuracy ± 0.05 percent of reading ± one digit

(5) Readout four digits minimum

(e) Oscilloscope with the following characteristics

(1) CRT 5 inches

(2) Input impedance 1 megohm with no more than a 47 picofarac shunt

(3) Bandwidth (6 centimeter reference signal) Zero to 20 MHz

(4) Risetime (6 centimeter reference signal): 18 nanoseconds maximum

(5) Sweep 0.1 microsecond per centimeter to 1 second per centimeter

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(6) Delay sweep. Variable from 0.2 microsecond per centimeter to 10 seconds per centimeter, accuracy, ± 3 percent, leading edge readable

(7) Vertical sensitivity 0.02 volt per centimeter to 10 volts per centimeter

(8) Horizontal sensitivity 0.1 volt per centimeter to 10 volts per centimeter

(9) Triggering Internal, external, and automatic

(f) Frequency Counter with the following characteristics

(1) Maximum counting rate. 1.0 MHz or greater

(2) Registration. 6 digits minimum

(3) Time base 1 MHz crystal oscillator

(4) Input sensitivity 0.1 volt rms

(5) Input impedance 1 megohm, 50 picofarads

(6) Frequency measurement 2 Hz to 1.0 MHz or greater

(7) Accuracy ± 1 count ± 2 x 10^{-4} percent

(g) Television Monitor with the following characteristics

(1) Size 17 inch screen (diagonal measurement)

(2) Resolution 800 lines minimum

(3) Video input impedance Resistive, 470 Kilohms minimum, capacitive, 15 picofarads maximum

(4) Video input signal Will operate with input of 0.25 volt to 4.00 volts peak-to-peak with negative sync pulses

(5) Horizontal scan rate 525 lines per second

4.5.2 Test Environment - Unless otherwise specified herein, tests shall be conducted under the following conditions

Temperature	Room ambient (25 ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity

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4.6 Preproduction inspections - The preproduction samples shall be subjected to the tests and examinations as specified in table 2. Failure of the preproduction samples to pass any of the required tests or examinations shall be cause for rejection of the samples.

TABLE II. Inspection requirements.

Inspection	Method Paragraph	Preproduction	Quality Conformance	Periodic Performance
Examination of Product	4.9.1	X	X	X
Dimensions and Weight	4.9.2	X		X
Voltage Measurement Tests	4.9.3.1	X	X	X
Internal Grounds	4.9.3.2	X	X	X
POWER Lamp, Front Panel Illumination, and HOURS Meter Test	4.9.3.3	X	X	X
Video Generator Tests	4.9.3.4	X	X	X
Electrical Dummy Load DA-514/ASM-184(V) Test	4.9.3.5	X	X	X
Reliability Index Determination	4.9.4	X		X
Reliability Assurance Tests	4.9.5		X	
Interference Control Tests	4.9.6	X		X
Environmental Tests	4.9.7	X		X
Illuminating Panel	4.9.8	X		X
Voltage Protection	4.9.9	X		X
Warm-Up Time	4.9.10	X		X

4.7 Quality Conformance Inspections - Each test set submitted for acceptance under the contract or order shall be submitted to the tests and inspections as specified in table II. Failure of any test set to pass any of the required tests or examinations shall be cause for rejection of that test set.

4.8 Periodic Production Inspections - The periodic production sample selected in accordance with 4.3.2 shall be subjected to the tests and examinations as specified in table II.

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4.9 Inspection Procedures - The following inspection procedures shall be followed in the performance of all tests and examinations required. Any diviation from these procedures shall be specifically approved by the procuring activity.

4.9.1 (M101) Examination of Product - The test set shall be examined to determine that all manufacturing, construction, and testing requirements of Data List DL 64A15F2002 have been met. In particular, the test set shall be examined for compliance to:

(a) The interchangeability requirements of 3.2.1.3.

(b) The nomenclature, nameplate, and marking requirements.

(c) The workmanship requirements.

(d) The construction requirements of 3.3.

(e) Packaging requirements of Section 5.

4.9.2 Diminsions and Weight - The test set and all accessory items shall be measured to determine that the overall weight of the complete test set and overall diminsions of the test set and accessory items conform to the requirements of 3.3.1, 3.3.2, and DL 64A15F2002.

4.9.3 (M102) Performance Tests - The test set shall be subjected as specified to the following performance tests.

4.9.3.1 Voltage Measurement Tests - Perform the following tests to determine that the test set is capable of measuring and registering the individual voltage quantities as applied to the pins of connector J1 and as supplied by the test set for self-checks. Set the FUNCTION switch to the positions indicated and depress the PUSH/TEST switch when taking all test readings.

4.9.3.1.1 Input Power Low Voltage Limit - With primary input power of 103.5 ± 2.5 volts rms, 400 ± 20 Hz applied to pins 23, 22, and 21 (7 neutral) of J1, and with the FUNCTION switch set to positions 1, 2, and 3, verify that the QUALITY meter registers over the left edge of the area containing a triangle.

4.9.3.1.2 Input Power High Voltage Limit - With primary input power of 126.5 ± 2.5 volts rms, 400 ± 20 Hz applied to pins 23, 22, and 21 (7 neutral) of J1, and with the FUNCTION switch set to positions 1, 2, and 3, verify that the QUALITY meter registers over the right edge of the area containing a triangle.

4.9.3.1.3 Input Power Phase Sequence - With primary input power of 115 ± 11.5 volts rms, 400 ± 20 Hz applied to pin 23 (phase A), pin 22 (phase B), pin 21 (phase C), and pin 7 (neutral) of J1, and with the FUNCTION switch set to position 4, verify that the QUALITY meter registers in the area containing a triangle. Interchange any two phases of the input power to the test set and verify that the QUALITY meter registers outside of and to the left of the area containing a triangle. Reconnect the input power in the proper phase sequence before continuing with the voltage measurement tests.

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4.9.3.1.4 28 Volt DC Signal Low Limit - With 23.8 ± 0.5 Volts dc applied between pins J1-15, J1-20, J1-19, J1-18, J1-11 or J1-10 (positive), and J1-7 (ground), and with the FUNCTION switch set to positions 5, 10, 11, 16 and 23, verify that the QUALITY meter registers over the left edge of the area containing a circle.

4.9.3.1.5 28 Volt DC Signal High Limit - With 32.2 ± 0.5 volts dc applied between pins J1-20, J1-19, J1-18, J1-11 or J1-10 (positive), and J1-7 (ground), and with the FUNCTION switch set to positions 5, 10, 11, and 23, verify that the QUALITY meter registers over the right edge of the area containing a circle.

4.9.3.1.6 28 Volt DC Power Supply - With the FUNCTION switch set to position 6, verify that the QUALITY meter registers within the area containing a circle.

4.9.3.1.7 Video Generator Circuits - With the FUNCTION switch set to positions 7, 8 and 9, verify that the QUALITY meter registers within the area containing a circle.

4.9.3.1.8 300 Volt DC Signal Low Limit - With 294 ± 2.0 volts dc applied between J1-17 (positive) and J1-7 (negative), and with the FUNCTION switch set to position 12, verify that the QUALITY meter registers over the left edge of the area designated by an "X".

4.9.3.1.9 300 Volt DC Signal High Limit - With 306 ± 2.0 volts dc applied between J1-17 (positive) and J1-7 (negative), and with the FUNCTION switch set to position 12, verify that the QUALITY meter registers over the right edge of the area designated by an "X".

4.9.3.1.10 Negative 300 Volt DC Signal Limits - With the polarity of the voltage reversed, repeat the tests of 4.9.3.1.8 and 4.9.3.1.9, and verify that the meter indications are as specified.

4.9.3.1.11 195 Volt DC Signal Low Limit - With 191 ± 1.5 volts dc applied between J1-17 (positive) and J1-7 (negative), and with the FUNCTION switch set to position 13, verify that the QUALITY meter registers over the left edge of the area designated by an "X".

4.9.3.1.12 195 Volt DC Signal High Limit - With 199 ± 1.5 volts dc applied between J1-17 (positive) and J1-7 (negative), and with the FUNCTION switch set to position 13, verify that the QUALITY meter registers over the right edge of the area designated by an "X".

4.9.3.1.13 Negative 195 Volt DC Signal Limits - With the polarity of the voltages reversed, repeat the tests of 4.9.3.1.11 and 4.9.3.1.12, and verify that the meter indications are as specified.

4.9.3.1.14 AC and Positive DC Stray Voltage Signal - With 0.100 (+0.010, -0.000) volt rms applied between J1-16 and J1-17 (positive), and J1-7 (ground), and with the FUNCTION switch set to positions 14 and 15, verify that the QUALITY meter registers in the BAD area immediately adjacent to the area containing a rectangle. Replace the a-c voltage with 0.100 (+0.010, -0.000) volt dc and verify that the QUALITY meter registers in the BAD area immediately adjacent to the area containing a rectangle.

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4.9.3.1.15 Negative DC Stray Voltage Signal - With a voltage of 0.100 (0.010, -0.000) volt dc applied between J1-15 and J1-17 (negative), and J1-7 (ground), and with the FUNCTION switch set to positions 21 and 22, verify that the QUALITY meter registers in the BAD area immediately adjacent to the area containing a rectangle.

4.9.3.1.16 Ground Resistance - With J1-4, J1-9, J1-13, and J1-14 connected to J1-7 by jumpers capable of passing 5.0 amps with a voltage drop no greater than 5 millivolts and with the FUNCTION switch set to positions 17, 18, 19, and 20, verify that the QUALITY meter registers in the center of the area containing a circle.

4.9.3.2 Internal Grounds - Perform the resistance measurements of 4.9.3.2.1 and 4.9.3.2.2 to determine that the resistance between the pins of connector J1 and ground are as specified.

4.9.3.2.1 Pins 5, 6, and 7 of J1 - With an ohmmeter connected between J1-5 and the front panel GND connector, and with the FUNCTION switch set in turn to each of its positions, verify that the resistance between J1-5 and ground is 0.5 ohm or less. Repeat the test with the ohmmeter connected between J1-6 and the front panel GND connector, then with the ohmmeter connected between J1-7 and the front panel GND connector.

4.9.3.2.2 Pin 30 of J1 - With the positive lead of an ohmmeter connected to J1-30 and the negative lead connected to the front panel GND connector, and with the FUNCTION switch set in turn to each of its positions, verify that the resistance between J1-30 and ground is as follows

FUNCTION Switch Position	Resistance Between J1-30 and Ground
OFF	5.0 megohms or greater
7 through 11	Less than 30 ohms
12 through 23	Less than 0.5 ohm

4.9.3.3 POWER Lamp, Front Panel Illumination, and HOURS Meter Test - With primary power of 115 ± 11.5 volts rms, 400 ± 20 Hz applied to pins 23, 22, and 21 (7 neutral), and with the POWER switch set to ON, verify that the POWER lamp and the front panel illuminate and the HOURS meter operates.

4.9.3.4 Video Generator Tests - With primary power of 115 ± 11.5 volts rms, 400 ± 20 Hz applied to pins 23, 22, and 21 (7 neutral) of J1, perform the tests of 4.9.3.4.1 through 4.9.3.4.5 to determine that the video generator characteristics are as specified.

4.9.3.4.1 Master Oscillator Frequency - With a frequency meter connected to TP4 of video assembly A3, and with the POWER switch set to ON, verify that the frequency of the master oscillator is 126.000 KHz ± 0.02 percent.

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4.9.3.4.2 Crossbar Pattern - With a television monitor connected across a 91-ohm, 1/2 watt or greater resistive load between J1-31 and ground, and with the POWER switch set to ON, verify that the crossbar pattern shown in figure 1 is displayed with a normal amplitude when the FUNCTION switch is set to position 1 and positions 12 through 22, and with an amplitude 50 ± 10 percent or normal amplitude when the FUNCTION switch is set to position 23. Each of the patterns shall have seven blocks in the vertical plane and five plus partial blocks on each side in the horizontal plane. The patterns shall also have straight black horizontal and vertical grid lines which separate each of the blocks one from another. Each of the blocks shall contain one shade of gray corresponding to a step in a logarithmic gray scale.

4.9.3.4.3 Stationary Band - With a television monitor connected across a 91-ohm, 1/2-watt resistive load between J1-31 and ground, with the POWER switch set to ON, and with the FUNCTION switch set to positions 2, 3, 4, 5, and 6, verify that stationary resolution lines, appearing as shown in figure 2 and having resolution as specified in table III, are displayed.

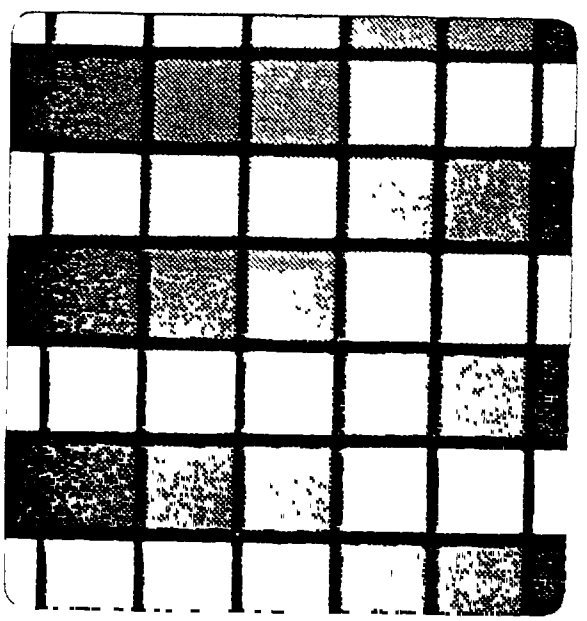


FIGURE 1. Crossbar pattern.

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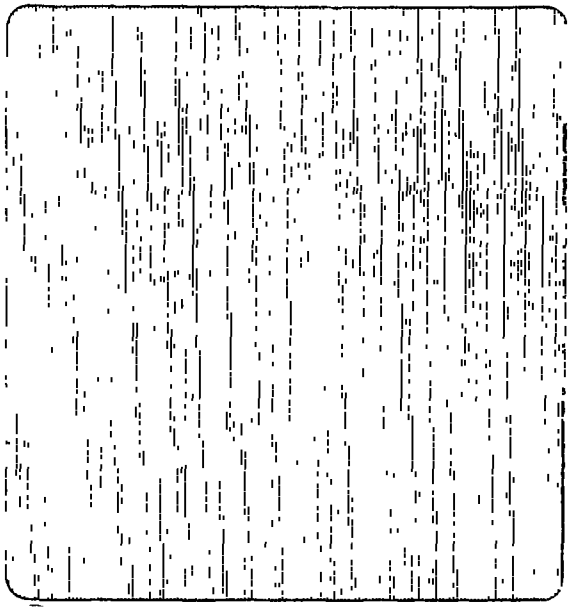


FIGURE 2. Stationary band television display.

TABLE III. Stationary band resolution.

FUNCTION Switch Position	Resolution
2	200 lines
3	300 lines
4	400 lines
5	500 lines
6	600 lines

4.9.3.4.4 Moving Band - With a television monitor connected across a 91-ohm resistive load between J1-31 and ground, and with the POWER switch set to ON and the FUNCTION switch set to OFF and positions 7 through 11, verify that a moving band of resolution lines, appearing as shown in figure 3 and having resolution as specified in table IV, is displayed.

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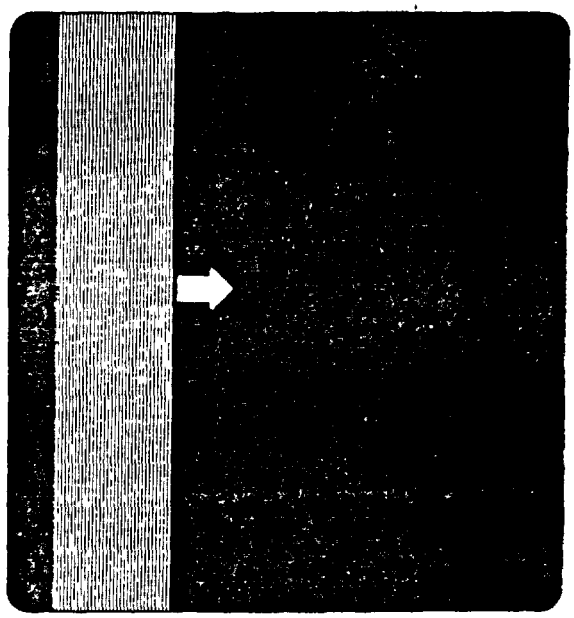


FIGURE 3. Moving band television display.

TABLE IV. Moving band resolution.

FUNCTION Switch Position	Resolution
7	200 lines
8	300 lines
9	400 lines
10	500 lines
11 and Off	600 lines

4.9.3.4.5 Composite Video - With an oscilloscope connected across a 91-ohm resistive load between J1-31 and ground, and with the POWER switch set to ON and the FUNCTION switch set as necessary to obtain the required synthetic video, verify that the composite video displayed on the oscilloscope meets the following requirements

(a) HORIZONTAL SYNC PULSE

(1) Repetition rate	15,750 Hz ± 0.02 percent
(2) Duration	8.0 (+1.0, -1.5) microseconds

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(3)	Blanking	16 (+1.0, -2.0) microseconds, including sync pulse
(4)	Amplitude	0.80 (+0.10, -0.25) volt peak-to-peak
(5)	Risetime and falltime	500 nanoseconds maximum
(6)	Polarity	Negative
(7)	Interlace accuracy	Pulses of successive fields to be displaced by 1/2 line time ± 10 percent
(8)	Droop	0.05 volt maximum
(9)	Overshoot and undershoot	0.08 volt maximum

(b) VERTICAL SYNC PULSE

(1)	Repetition rate	60 Hz ± 0.02 percent
(2)	Duration	189 ± 19 microseconds
(3)	Blanking	
	[a] Rest (front porch)	Equal to three horizontal lines ± 10 percent. There shall be no horizontal sync pulses during front porch.
	[b] Blanking (back porch)	Equal to 10 horizontal lines ± 10 percent. Horizontal sync pulses to be generated during this interval.
(4)	Amplitude	0.80 (+0.10, -0.25) volt peak-to-peak
(5)	Risetime and falltime	1 microsecond maximum
(6)	Polarity	Negative
(7)	Droop	0.10 volt maximum
(8)	Overshoot and undershoot	0.08 volt maximum

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(c) SYNTHETIC VIDEO, MOVING BAND

(1) Amplitude	2.20 ± 0.50 volts peak-to-peak (black to white)
(2) Polarity	Positive
(3) Moving Band	
[a] Width	10 ± 2 microseconds
[b] Start	18 (+2, -1) microseconds from the leading edge of the horizontal sync pulses
[c] Run length	31 ± 2.5 microseconds
[d] Run duration	2 ± 0.3 seconds
(4) Pulse overshoot	0.01 volt maximum

(d) SYNTHETIC VIDEO, CROSSBAR

(1) Amplitude	2.20 ± 0.50 volts peak-to-peak (blak to white level)
(2) Polarity	Positive (white level)
(3) Voltage steps	7
(4) Voltage difference (between successive steps	0.3 ± 0.1 volt
(5) Voltage level, first step, ascending staircase	0.125 ± 0.075 volt above black level
(6) Step voltage rise and falltime	400 nanoseconds maximum
(7) Step voltage overshoot	0.08 volt maximum into the sync and 0.12 volt maximum in the vertical direction
(8) Step voltage droop	0.05 volt maximum

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(9) Vertical grid line pulse		
	[a] Duration	1.0 ± 0.25 microsecond
	[b] Overshoot	0.08 volt maximum into sync and 0.12 maximum in vertical direction
	[c] Risetime and falltime	400 nanoseconds maximum
(10) Horizontal grid line pulses		
	[a] Duration	260 (+30, -60) microseconds
	[b] Overshoot	0.08 volt maximum
(e) SYNTHETIC VIDEO, RESOLUTION LINES		
(1) Amplitude		2.20 ± 0.50 volts peak-to-peak (black to white level)
(2) Polarity		Positive
(3) Resolution Lines		
	[a] 200 lines	2.0 MHz ± 10 percent
	[b] 300 lines	3.0 MHz ± 10 percent
	[c] 400 lines	4.0 MHz ± 10 percent
	[d] 500 lines	5.0 MHz ± 8 percent
	[e] 600 lines	6.0 MHz ± 8 percent
(4) Pulse overshoot		0.08 volt maximum

4.9.3.5 Electrical Dummy Load DA-514/ASM-184(V) Test - With the test equipment connected as shown in figure 4, and with circuit breaker CB1 set to ON, verify that DS1 lights and the voltage across the 5-ohm resistor drops to 0.2 volt or less within 1 second after closing switch S1. With the test setup of figure 4 connected to J2 instead of J1, and with circuit breaker CB2 set to ON, verify that DS2 lights and the voltage across the 5-ohm resistor drops to 0.2 volt or less within 1 second after closing switch S1.

4.9.4 (M103) Reliability Index Determination - Test sets shall be tested for reliability index determination in accordance with MIL-R-22973, Test Level II.

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4.9.4.1 Length of Tests - The total unit hours of test, not including any burn-in period, shall be 1400 hours or until 10 failures have occurred whichever is first.

4.9.4.2 Duty Cycle - The equipment shall be operated on a continuous duty cycle.

4.9.4.3 Performance Characteristics to be Measured - During the reliability test, the tests specified in the paragraphs listed shall be performed an average of once every 24 hours of test set "ON" time

4.9.3.1.1	4.9.3.1.14
4.9.3.1.3	4.9.3.1.16
4.9.3.1.4	4.9.3.4.2
4.9.3.1.6	4.9.3.4.3
4.9.3.1.7	4.9.3.4.4
4.9.3.1.8	

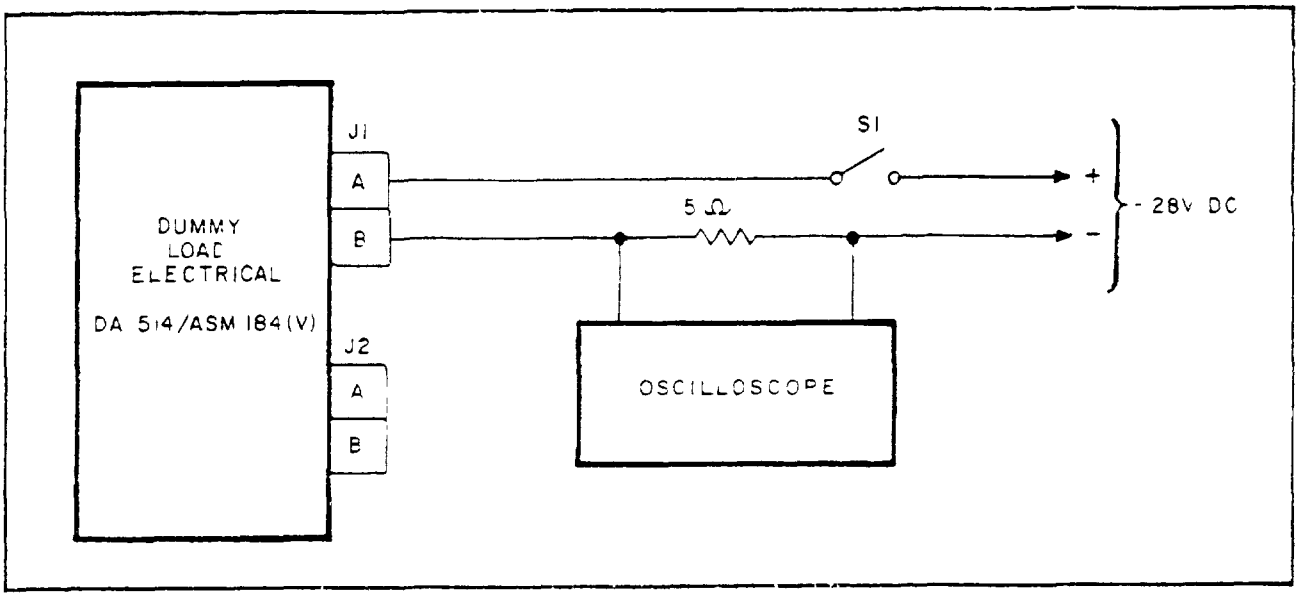


Figure 4. Electrical dummy load test setup.

4.9.4.4 Failure Criteria - In addition to the requirements as specified in MIL-R-22973, a failure shall have occurred whenever the performance characteristics of the test set fall outside of the requirements of Section 3. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless one caused the other parts to fail.

4.9.5 (M104) Reliability Assurance Tests - Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests shall be conducted as required for both the Qualification Phase and Sampling Phase.

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4.9.5.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of two equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be in accordance with MIL-STD-781, table 5. For the qualification phase, Test Level B shall be used with the exception that the equipment shall operate continuously. The accept-reject criteria for Test Plan III shall be used.

4.9.5.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as specified in MIL-STD-781 under the section entitled "Sampling Phase of Production Reliability Tests". For the sampling phase, Test Level B shall be used, except that the equipment shall operate continuously. The accept-reject criteria for Test Plan V shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.9.5.3 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified 4.9.4.3.

4.9.5.4 Equipment Failure - Equipment failure shall be as specified in 4.9.4.4.

4.9.5.5 Allowed Maintenance During Reliability Tests - During the period of the tests, no preventive maintenance measures except readjustment of circuits having variable controls shall be performed on the equipment. The stability requirements as specified in 3.3.3.1 apply.

4.9.6 (M105) Interference Control Tests - The test set and its components shall be subjected to the interference and susceptibility tests specified in MIL-I-6181 with limits modified as specified in 3.3.4.

4.9.7 (M106) Environmental Tests - Environmental tests shall be conducted as specified in MIL-T-5422, excluding the fungus test and the sand and dust test, and modifying the limits of the applicable tests to correspond with the limits specified herein. In addition, the test set combination case shall be subjected to the drop and watertightness test as specified in MIL-T-21200.

4.9.8 Illuminating Panel - The illuminating panel shall be examined for legibility of all markings. In addition the panel shall be subjected to the illumination tests as specified in MIL-P-7788.

4.9.9 (M107) Voltage Protection - After being subjected to the transient voltages specified in MIL-STD-704, figure 2, the test set shall be examined for any damage which may have occurred. After the input power to the test set returns to that specified in 3.3.8, the test set shall be tested for normal operation by placing the FUNCTION switch in positions 6, 7, 8, and 9 and observing the QUALITY meter for the normal indications specified in 3.5.1.1.6 through 3.5.1.1.9.

4.9.10 (M108) Warm-Up Time - During the following tests the equipment shall be considered to be in an operational condition when the requirements as specified in 4.9.3.1.3, 4.9.3.1.6, and 4.9.3.4.2 have been met.

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4.9.10.1 Warm-Up Time (Standard Conditions) - With the test set supplied input power as specified 3.3.8, verify that the equipment has warmed-up to an operational condition one minute after the POWER switch is placed in the ON position.

4.9.10.2 Warm-Up Time (Extreme Conditions) - With the temperature of the test set stabilized at a temperature of -40°C and with the test set supplied input power as specified in 3.3.8, verify that the equipment has warmed-up to an operational condition within five minutes after the POWER switch is placed in the ON position.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - These portable test sets are intended for hanger deck or flight line use to conduct operational and safety checks on aircraft wiring and video circuits associated with the WALLEYE Guided Weapon.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply

(a) Contract - The contract shall have precedence over any specification.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) The Drawings of DL 64A15F2002 - The drawings shall have precedence over other referenced specifications. Any deviation from the drawings shall be specifically approved by the procuring activity.

(d) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.3 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

(c) Data requirements (see 3.6).

6.4 This specification is under the cognizance of AIR 549321.

Custodian.
Navy-AS

Preparing Activity NAVY-AS
Project No. 6625-N755

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE SUBMISSION *(YYMMDD)*

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